

Yixiang Gao, PhD, *Electrical and Computer Engineering*

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PROFESSIONAL SUMMARY

Machine learning and robotics researcher. Exhibits profound expertise in **Deep Learning, Computer Vision, Robotics and Pattern Recognition**. Specialized in fostering interdisciplinary collaborations, notably with speech language pathology, and underground mining related applications. Advocate for open source technologies. Seeking to leverage extensive AI/ML expertise to address complex real-world challenges and drive innovation in technology application.

EDUCATION

University of Missouri - Columbia	Ph.D in Electrical and Computer Science	2017 – 2024
University of Missouri - Columbia	BS in Electrical Engineering	2014 – 2017
University of Missouri - Columbia	BS in Computer Engineering	2014 – 2017

EXPERIENCE

Post Doctorial Fellow | *Missouri S&T | Department of Mining & Explosives Engineering* Oct 2024 – Current

- Robotics researcher for CDC NIOSH Initiative to empower miners for self-escape
 - Lead research on enhancing deep learning-based object detection algorithms such as YOLO in mine safety applications.
 - Integrate advanced LiDAR and thermal camera on Boston Dynamics’ Spot quadrupled robot with ROS2.
 - Develop autonomous navigation solutions with Spot for underground miner exploration and miner safety monitoring.

Graduate Research Assistant | *University of Missouri - Columbia | ViGIR Lab* Aug 2017 – Sep 2024

- PhD Thesis: Confounded predictions in machine learning
 - Detect, quantify, and mitigate confounding factors in machine learning and deep learning models.
- Student Investigator for NIH R01DC018026 Aug 2019 – Sep 2024
 - Developed classification pipeline with transformers and CNN using Pytorch and Matlab for analyzing both sEMG and fMRI data.
- Student Investigator for NIH R15DC015335 Aug 2017 – Aug 2019
 - Developed machine learning algorithms with SVM, K-means, and MLP using scikit-learn and Pytorch for analyzing sEMG data

JOURNAL PUBLICATIONS

[1] M. Défaza, K. Awuah-Offei, and **Yixiang Gao**, “Stall detection in hydraulic excavator operations using heuristics and machine learning: A case study,” *Automation in Construction*, 2025 (Under Review).

[2] M. Berardi, E. Tippit, **Yixiang Gao**, G. N. DeSouza, *et al.*, “Automated analysis of relative fundamental frequency in continuous speech: Development and comparison of three processing pipelines,” *Journal of Voice*, 2025. doi: <https://doi.org/10.1016/j.jvoice.2025.04.006>.

[3] **Yixiang, Gao**, M. Dietrich, and G. N. DeSouza, “Classification of vocal fatigue using sEMG: Data imbalance, normalization, and the role of vocal fatigue index scores,” in *Applied Sciences*, 2021. doi: 10.3390/app11104335.

PEER-REVIEWED CONFERENCES

[1] C. Addy, A. Gurumadaiah, **Yixiang Gao**, and K. Awuah-Offei, “A comprehensive dataset for underground miner detection in diverse scenario,” in *20th International Symposium on Visual Computing*, 2025 (Under Review).

[2] **Yixiang Gao** and K. Awuah-Offei, “Navigation in underground mine environments: A simulation framework for quadruped robots,” in *2025 IEEE 21st International Conference on Automation Science and Engineering*, 2025 (Accepted).

[3] R. Farag, P. C. Upadhyay, J. Demby’s, **Yixiang Gao**, *et al.*, “Efficientnet-sam: A novel effecientnet with spatial attention mechanism for covid-19 detection in pulmonary ct scans,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2024. doi: 10.1109/CVPRW63382.2024.00528.

[4] **Yixiang, Gao**, M. Berardi, M. Dietrich, and G. N. DeSouza, “Removal of confounding factors using GA-SVM feature adaptation: Application on detection of vocal fatigue thru semg classification,” in *IEEE 2023 Congress on Evolutionary Computation (CEC)*, 2023. doi: 10.1109/CEC53210.2023.10253983.

- [5] J. Demby's, **Yixiang, Gao**, and G. N. DeSouza, "A study on solving the inverse kinematics of serial robots using artificial neural network and fuzzy neural network," in *2019 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, 2019. doi: 10.1109/FUZZ-IEEE.2019.8858872.
- [6] J. Demby's, **Yixiang, Gao**, A. Shafiekhani, and G. N. DeSouza, "Object detection and pose estimation using CNN in embedded hardware for assistive technology," in *2019 IEEE Symposium Series on Computational Intelligence (SSCI)*, 2019. doi: 10.1109/SSCI44817.2019.9002767.
- [7] **Yixiang, Gao**, M. Dietrich, M. Pfeiffer, and G. N. DeSouza, "Classification of sEMG signals for the detection of vocal fatigue based on VFI scores," in *2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2018. doi: 10.1109/EMBC.2018.8513224.

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- [1] R. Farag, P. C. Upadhyay, **Yixiang Gao**, J. Demby's, *et al.*, "Covid-19 detection from pulmonary ct scans using a novel efficientnet with attention mechanism," *CoRR*, 2024.

CONFERENCE PRESENTATION

- [1] **Yixiang, Gao**, G. N. DeSouza, M. Berardi, and M. Dietrich, "Voice sEMG classification of sentences for vocal fatigue detection using GA-SVM for confounder," in *13th International Conference on Voice Physiology and Biomechanics (ICVPB)*, 2024.
- [2] **Yixiang, Gao**, M. Berardi, M. Dietrich, and G. N. DeSouza, "Feature adaptation with GA-SVM for confounding removal - an application on vocal fatigue detection using semg classification," in *The 15th Advances in Quantitative Laryngology, Voice and Speech Research (AQL)*, 2023.
- [3] **Yixiang, Gao**, M. Dietrich, and G. N. DeSouza, "Classification of vocal fatigue using neck sEMG with leave-one-subject-out testing," in *The 14th Advances in Quantitative Laryngology, Voice and Speech Research (AQL)*, 2021.
- [4] **Yixiang, Gao**, M. Dietrich, and G. N. DeSouza, "Explore voice production variability through neck semg clustering - challenge for accurate labeling of vocal fatigue," in *The 14th Advances in Quantitative Laryngology, Voice and Speech Research (AQL)*, 2021.
- [5] M. Dietrich, E. Tippit, A. Walker, M. Pfeiffer, *et al.*, "Relative fundamental frequency during vocal loading and relationships with laryngeal muscular patterns," in *The 13th Pan-European Voice Conference*, 2019.
- [6] M. Dietrich, **Yixiang, Gao**, and G. N. DeSouza, "Extralaryngeal surface emg features that distinguish between those with and without elevated scores on the vocal fatigue index," in *The Fall Voice Conference*, 2019.
- [7] **Yixiang, Gao**, M. Dietrich, M. Pfeiffer, A. Walker, *et al.*, "Classification of vocal gestures extracted from quasi-daily sentences to detect vocal fatigue," in *The 13th Advances in Quantitative Laryngology, Voice and Speech Research (AQL)*, 2019.
- [8] **Yixiang, Gao**, M. Pfeiffer, M. Dietrich, and G. N. DeSouza, "Classification of neck surface EMG signals for the early detection of vocal dysfunction," in *The 12th Advances in Quantitative Laryngology, Voice and Speech Research (AQL)*, 2017.

TECHNICAL SKILLS

Programming Languages: Python, C/C++/CUDA, Matlab, HTML
Libraries: pytorch, scikit-learn, ROS 2, Isaac Sim
MLOps and API Tools: Git, Docker, Kubernetes, Github Actions for CI/CD
Operating Systems: Linux(preferred), Windows
Languages: English, Chinese

OTHER ACTIVITIES

Academic Paper Reviews: EMBC2019/2020/2021, CEC2023, CASE2025, Applied Sciences, TAI
Open-Source Project Contributions: tinygrad, huggingface transformers